

Honest limits. At $k = 5$ the threshold 162 is *not* optimal: $\ell = 73$ and $76 \leq \ell \leq 107$ are still unsolved, and filling them would bring it down to 54. $k \geq 6$ has no tiles at all. $Z(P(n, 4))$ is known for every n , but the *matrix* value $M(P(n, 4))$ is still open at the single value $n = 24$, where $Z = 10$ but no certificate exists. The $k \geq 6$ congruence-class values are numerical, a weaker standard than the rest. The rank law is verified over a range, not proved.

What we cannot diagnose. We first searched $k = 5$ over $41 \leq \ell \leq 45$ —